Learning to Count Words in Fluent Speech enables Online Speech Recognition

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Abstract

Sequence to Sequence models, in particular the Transformer, achieve state of the art results in Automatic Speech Recognition. Practical usage is however limited to cases where full utterance latency is acceptable. In this work we introduce Taris, a Transformer-based online speech recognition system aided by an auxiliary task of incremental word counting. We use the cumulative word sum to dynamically segment speech and enable its eager decoding into words. Experiments performed on the LRS2 and LibriSpeech datasets, of unconstrained and read speech respectively, show that the online system performs on a par with the offline one, while having a dynamic algorithmic delay of 5 segments. Furthermore, we show that the estimated segment length distribution resembles the word length distribution obtained with forced alignment, although our system does not require an exact segment-to-word equivalence. Taris introduces a negligible overhead compared to a standard Transformer, while the local relationship modelling between inputs and outputs grants invariance to sequence length by design.

1 Introduction

Having a natural conversation with a computer has fascinated humankind for a long time. A key ingredient of this ambition is granting computers the ability to recognise spoken words with minimum latency. This allows a more interactive communication, where the computer is able to interrupt a speaker to acknowledge or ask for clarifications.

Despite the remarkable progress in end-to-end automatic speech recognition technology based on sequence to sequence neural network architectures [8], an unresolved issue is reducing the latency from full utterances down to a few words. This sentence-level, or offline conditioning, is a fundamental barrier to online decoding.

Humans develop the ability to segment words in continuous speech from the earliest stages of life [19]. There is evidence that we integrate a set of acoustic, phonetic, prosodic, and statistical cues in order to segment words in fluent speech [18]. This leads us to ask whether the ability to segment speech into *word* units with a neural network offers the potential to help crack the challenge of decoding online. This approach would take advantage of the monotonicity of speech, allow the network focus on local properties, and remove the offline conditioning.

To this end, we introduce Taris [1], a Transformer-based system for online speech recognition that learns to model the local relationships between text and audio in speech, relaxing the global conditioning constraint of the original model. We achieve this through self-supervision by introducing an auxiliary

[1] https://github.com/georgesterpu/Taris

Preprint. Under review.

word counting task which facilitates the segmentation of speech. Taris allows efficient minibatch training and introduces a negligible overhead compared to the original Transformer model, without trading off the recognition accuracy. We make our implementation publicly available.

2 Background

A major technical challenge in learning to segment speech is the difficulty of formulating the problem in a fully diffentiable framework. Some previous attempts include the Recurrent Neural Network Transducer [15, 14, 31, 3], Neural Transducer [17, 32], segmental conditional random fields [5, 36], hard monotonic attention [24, 30], segment attention [9, 13, 16], or triggered attention [26]. However the models made use of dynamic programming, training in expectation, or policy gradients, leading to training difficulties. Our work retains the segment attention design, but approaches the problem of speech segmentation from a different angle. By learning to count words through self-supervision, we introduce a mechanism that allows end-to-end training using only backpropagation.

More recent proposals in online speech recognition address this challenge by assuming one sub-word unit per segment [23, 11], or discover an inventory of sub-word units [12], a concept previously explored in machine translation [20]. Our focus in this work is on word units. In English, words allow a monotonic and bijective mapping between their acoustic and symbolic representations, however these properties do not hold at the sub-word level due to the highly complex spelling rules in English orthography. Moreover, words can be counted in a deterministic way, which allows us to introduce a self-supervision word counting task without requiring new annotations.

The sequence to sequence (seq2seq) architecture was proposed in [2, 35]. An Encoder transforms a variable length input sequence into a sequence of latent representations, and a Decoder maps the latent sequence onto a target sequence of a different length, aiming to establish a soft-alignment between elements of the inputs and the targets. In attention-based seq2seq networks, the conditional dependency of each output token on the entire input sequence prohibits online decoding. Yet, it has been shown that, once convergence is reached, there are predominantly local relationships between the output tokens and the audio representations in speech [10, 6]. Therefore, potentially incurring no loss in accuracy, a local conditioning of the outputs on the inputs would break the offline limitation and reduce the algorithmic latency. The new challenge is to learn robust associations between input and output substrings which stand for the same linguistic concepts.

The Transformer [38] is a good seq2seq candidate for this task and we choose it as a foundation for our system Taris. Unlike the recurrent neural network that uses causal connections between timesteps (Figure 1a), the Transformer allows feature contextualisation at the sequence level through self-attention, illustrated in Figure 1b. This offline modelling strategy provides a theoretical upper limit of the segmentation performance. Furthermore, the self-attention connections in the Transformer block can be adjusted to allow causal modelling (Figure 1c) or non-causal modelling with a window (Figure 1d). The window length is directly linked to the algorithmic latency of Taris and its accuracy, and we investigate this trade-off in Section 5.3.

3 Model architecture

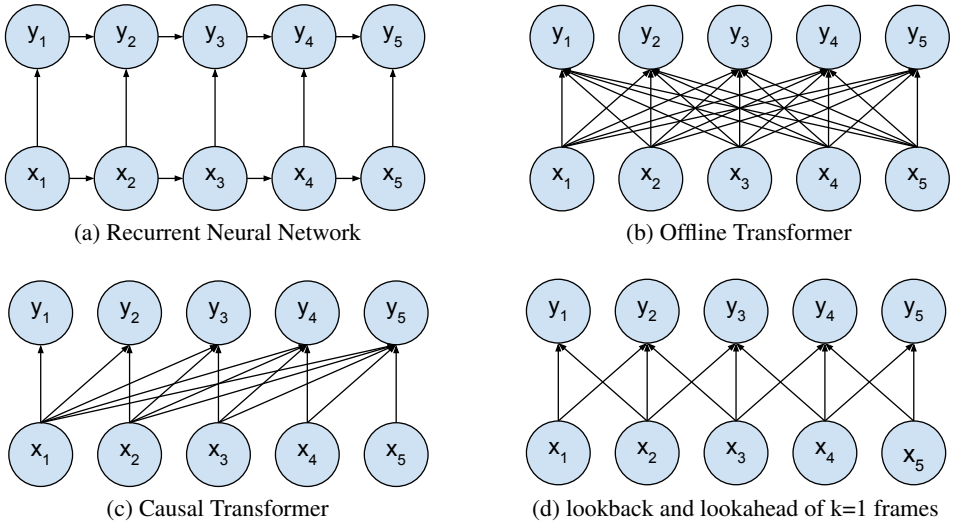
3.1 Encoding

Taris takes as input a variable length sequence of audio vectors $\mathbf{a} = \{a_1, a_2, \ldots, a_\mathbf{N}\}$ and applies the Encoder stack of the Transformer model defined in [38]. Because of latency considerations, instead of the original full connectivity in Figure 1b, we use the type displayed in Figure 1d, with controlled look-back e_{LB} and look-ahead e_{LA} frames. We denote the outputs of the encoder as:

$$\mathbf{o_A} = \mathsf{Encode}(\mathbf{a}, e_{LB}, e_{LA}) \tag{1}$$

Next, we apply a sigmoidal gating unit on each encoder output o_{A_i} to obtain a scalar score for each frame:

$$\alpha_i = \mathsf{sigmoid}(o_{A_i} W_G + b_G) \tag{2}$$
$$\text{where } \mathsf{sigmoid}(x) = \frac{1}{1+\exp(-x)}, W_G \in \mathbb{R}^{\mathrm{h} \times 1}, b_G \in \mathbb{R}^1$$



(a) Recurrent Neural Network (b) Offline Transformer

(c) Causal Transformer (d) lookback and lookahead of k=1 frames

Figure 1: Typical connectivity patterns at the sequence level for representation learning

We assign every single input frame i to a segment index $\hat{w}_i$ by taking the *cumulative sum* of α and applying the *floor* function on the output:

$$\hat{w}_i = \left\lfloor \sum_{j=1}^{i} \alpha_j \right\rfloor \tag{3}$$

Namely, the first predicted segment is delimited by a cumulative sum of α between 0 and 1, the second segment by the same quantity between 1 and 2, and so on.

3.2 Decoding

During training, the Decoder stack receives the labelled grapheme sequence $\mathbf{y} = \{y_1, y_2, \ldots, y_{\mathbf{L}}\}$, made of English letters and the unique word delimiter SPACE. We assign every grapheme k to a word index w_k by leveraging the SPACE tokens in the labelled sequence:

$$w_k = \sum_{j=1}^{k} (y_j == \mathsf{SPACE}) \tag{4}$$

Thus, whereas symbolic segmentation of speech uses a unique SPACE token to separate words, acoustic segmentation flags word boundaries by tracking the frame locations where the partial sum of the word counting signal α_i passes to the next integer value.

We modify the decoder-encoder connectivity of the Attention layer of [38] to allow our decoder to perform soft-alignment over a *dynamic* window of segments estimated by the encoder. More precisely, we only allow those connections for which the following condition is met:

$$\mathbf{V} = \widehat{W}_{ik} \leq (W_{ik} + d_{LA}) \textbf{ and } \widehat{W}_{ik} \geq (W_{ik} - d_{LB}) \tag{5}$$

In (5), d_{LA} and d_{LB} denote the number of segments the decoder is allowed to look-ahead and look-back respectively. The W and $\widehat{W}$ matrices are obtained from the w and $\hat{w}$ arrays by applying the tile operation, which repeats one sequence for a number of times equal to the length of the other one. In more detail, $\mathbf{V}$ is a 2D matrix $\in \mathbb{R}^{\mathrm{N} \times \mathrm{L}}$ that defines the admissible connections between any decoder timestep and any encoder timestep, acting as a bias on the decoder-encoder attention . The extension to 3D tensors that include the batch dimension is straightforward, offering Taris efficient minibatch training and inference.

The decoder implements a traditional character level auto-regressive language model that predicts the next grapheme in the sequence conditioned on all the previous characters and the dynamic audio context vector c_k:

$$c_k = \mathsf{Attention}(\mathsf{o}_{\mathsf{A}}^{\mathsf{valid}}, o_{D_{k-1}}) \tag{6}$$

$$o_{D_k} = \mathsf{Decode}(y, c_k) \tag{7}$$

$$p_k \equiv P(y_k | c_k, y_{1:k-1}) = \mathsf{softmax}(o_{D_k} W_v + b_v) \tag{8}$$

$$\text{where } W_v \in \mathbb{R}^{\mathrm{h \times v}}, b_v \in \mathbb{R}^{\mathrm{v}}$$

In (8), v is the vocabulary size of 28 tokens. We measure the difference between the estimated word sum $\Sigma\hat{w} = \sum_i \alpha_i$ and the true word count $|w| = \sum_k (y_k == \mathsf{SPACE})$ as:

$$Word\ Loss = (|w| - \Sigma\hat{w})^2 \tag{9}$$

We define the training loss as:

$$CE\ Loss = \frac{1}{L} \sum_k -y_k \log(p_k) \tag{10}$$

$$\mathsf{Loss} = CE\ Loss + \lambda\, Word\ Loss \tag{11}$$

In all our experiments we used a scale factor $\lambda = 0.01$ found empirically. The self attention connections of the Decoder are causal as in Figure 1c, since the model has to be auto-regressive.

Taris requires a negligible overhead in parameters (given by the W_G and b_G variables in (2)) and operations (equations (2)-(5)) over the original Transformer.

4 Why learn to count

Proper lexical segmentation of speech depends on context and semantics, as commonly illustrated by the example *how to wreck a nice beach* sounding similar to *how to recognise speech*. Thus, strategies incrementally scanning for hard boundaries [23, 16, 11] are less suited to word units, prompting [11] to perform beam search on the entire sequence of sub-word tokens estimated from each segment. Instead, Taris has to develop intrinsic word counting mechanisms. One plausible strategy is to incrementally gather lexical evidence at the sub-word level, and learn to represent boundary-informative acoustic cues on a manifold where they can be accumulated.

We conjecture that learning the ability to count words must facilitate the segmentation of speech into words, and we discuss below our intuition behind it.

In Figure 2 we illustrate the word counting sub-problem to be solved by the network. Starting in the bottom left corner, the network predicts scores for every audio frame in the sentence, and their sum must get as close as possible to the total word count, shown with a red circle. There is a very large number of paths that can be taken to reach the target count. However, when trained on large amounts of naturally distributed speech, we expect that Taris converges towards genuine word segmentation by having the cumulative sum cross all the intermediate word boundaries shown with yellow circles. In other words, the network should learn to self-normalise the accumulated probabilities for each word regardless of their length or cued structure.

We believe it suffices to train a system with the right amount of speech data, with the following intuition. As words appear in multiple contexts throughout a dataset, learning to count words should then have a normalisation effect on the fraction of $\Sigma\hat{w}$ allocated to each word in a sentence. Each word unit will approach a unitary mass allocation as its acoustic realisation is seen more often in multiple contexts. For the less frequent words, the correct allocation may happen by marginalisation if the sentences they appear in contain relatively more frequent words. Loosely speaking, it is the task of solving a system of linear equations where the variables are the partial sums corresponding to the acoustic frames between two consecutive estimated boundaries.

Since we do not explicitly model the pauses between words, and the convergence towards the segmental behaviour is a mathematical conjecture without analytic proof for now, it is likely to observe deviations in practice on learnt solutions. However, Taris does not require a very strict approximation of word boundaries to function correctly. Instead, it is sufficient to just avoid frequent under- and over-segmentation, as it directly impacts the model's latency.

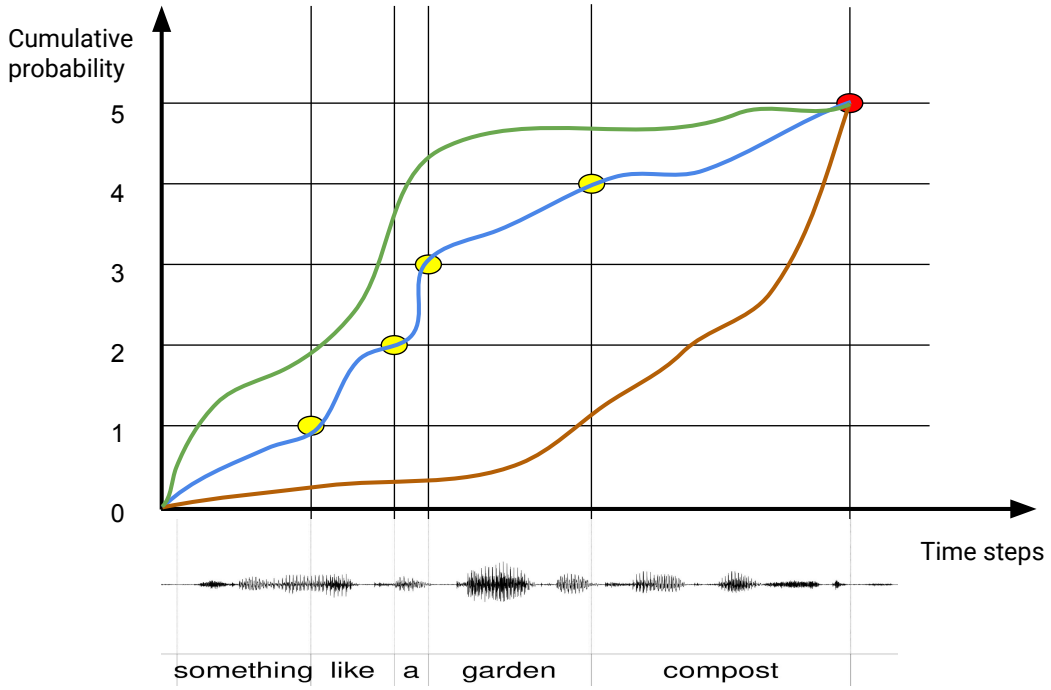


Figure 2: Illustration of the word counting problem. Given a spoken phrase *something like a garden compost*, the network must reach the correct word count 5 taking any path starting in the bottom left corner. The blue path allows an easy segmentation of speech into words. The green and dark orange paths are possible, but do not facilitate segmentation.

5 Experiments and Results

We first conduct our experiments on the audio part of the unconstrained speech dataset LRS2 [4] for rapid prototyping, and on the 100h partition of LibriSpeech [28] for empirical validation at a larger scale. To extract audio features $\mathbf{a}$ in Equation 1, we apply the log scale Short-time Fourier Transform on the waveform inputs, following same procedure as in [34] for noise corruption at 10, 0, and -5 db.

Our implementation of Taris is forked from the official Transformer model in TensorFlow 2 [37]. We train our LibriSpeech models for a total of 500 epochs at an initial learning rate of 0.001, decayed to 0.0001 after 400 epochs. The training time is approximately 200 seconds for a single epoch of LibriSpeech 100h on an Nvidia Titan XP GPU. The LRS2 models were trained with the same learning rates for 100 and 20 epochs respectively, on each noise level.

5.1 Neural network details

Our models use 6 layers in the Encoder and Decoder stacks, a hidden model size $d_{model} \equiv h = 256$, a filter size $d_{ff} = 256$, one attention head, and 0.1 dropout on all attention weights and feedforward activations. The models occupy 25 MB on disk.

5.2 The End-of-sentence token

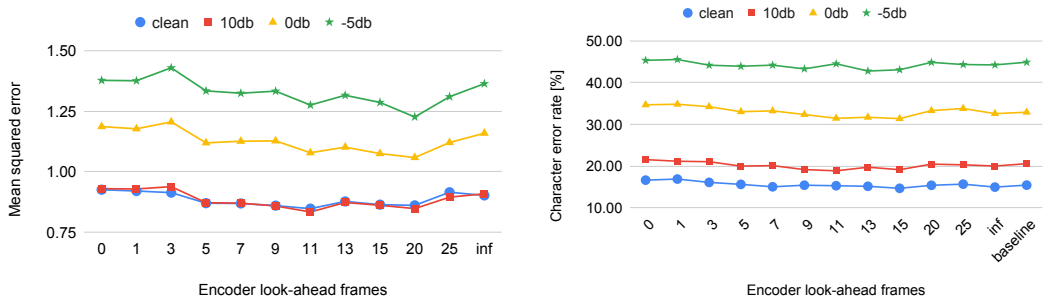
During our initial experiments, we noticed that traditional evaluation and training strategies in neural speech recognition are commonly misusing the End-of-sentence (EOS) token, making it difficult to evaluate online systems. The commonly used ASR datasets are a collection of variable-length utterances, and the system's accuracy is computed for each utterance using an edit distance based algorithm. These utterances are often fragments from full spoken sentences, such as the one illustrated in Figure 2, that were cropped using voice activity detection algorithms (e.g. in LRS2), and sometimes the fragmentation includes the ending and the start of two consecutive sentences, with the punctuation removed from the ground truth transcription (e.g. in LibriSpeech). In other words, the ASR system

does not receive full sentence units, and cannot develop the linguistic notion of an end of sentence. In our experiments it became obvious that one way the ASR model differentiates between an EOS token and a word delimiter (SPACE) likely comes from the apriori knowledge of the sentence length, and that EOS becomes more likely as the decoder-encoder alignment distribution advances towards the last remaining audio frames in the sentence.

The aspect above becomes problematic in an online setting, as the decoder is fed with a limited acoustic context. Given the nature of the dataset utterances, an online decoder does not know when to stop the decoding process, as EOS cannot be estimated even spuriously anymore. Online decoding would often stop after just a few words in an utterance, biasing the performance on longer sentences.

To circumvent this problem, we made two important changes to the traditional model. First, we replaced the EOS token in the labels, which cannot be predicted reliably, with the SPACE token. Second, we modified the stopping condition of the beam search inference decoder as follows: instead of stopping when all beams reach the EOS token, it now stops when the decoder predicts as many words as there were estimated by the audio encoder. This new strategy is mostly beneficial to the evaluation procedure, but should also be useful in practice as it allows the decoder to emit a controllable number of words. With this change, we are able to evaluate the error rate of Taris on full test sentences for which we lack any label alignments.

5.3 Learning to count words



(a) Word Loss on LRS2

(b) Character Error Rate on LRS2

Figure 3: Offline system evaluation for an increasing length of feature contextualisation in the encoder

We first investigate to what extent a sequence to sequence Transformer model can learn to count the number of words from audio data on LRS2. We train multiple models and gradually increase the number of encoder look-ahead frames e_{LA} to measure the variation of the $Word\ Loss$ as more future context becomes available. We see in Figure 3a that the mean squared word count error is sub-unitary in clean speech and 10db noise, i.e. the estimated count is less than one word away from truth. This suggests that words can be counted relatively well from acoustic speech. In addition, using a future context length of 11 frames, or 330ms in our setup, offers the best counting performance under all noise conditions. In Figure 3b we plot the mean Character Error Rate achieved by all our systems, including the offline Transformer baseline without the auxiliary Word Loss, and we observe no significant difference, with the 95% confidence intervals of the mean errors between 1% and 1.4%. Therefore, this auxiliary task is not detrimental to the original performance on LRS2.

5.4 Online ASR decoding

The decoder in our previous experiment had access to the entire encoder memory. For our online model we opt for an encoder lookahead e_{LA} of 11 frames and infinite lookback $e_{LB} = \infty$, as we showed in Section 5.3 that there are diminishing gains beyond this threshold. This roughly corresponds to an encoding latency of 330 msec.

In this experiment we evaluate the performance of Taris on LRS2 for an increasing number of decoder look-ahead segments d_{LA}, while setting the look-back value $d_{LB} = \infty$. For a practical online model it may be a good trade-off to limit the decoder look-back context to a single sentence when

transcribing continuously. We plot the Character Error Rate in Figure 4 for an increasing number of acoustic segments that the decoder is allowed to attend to.

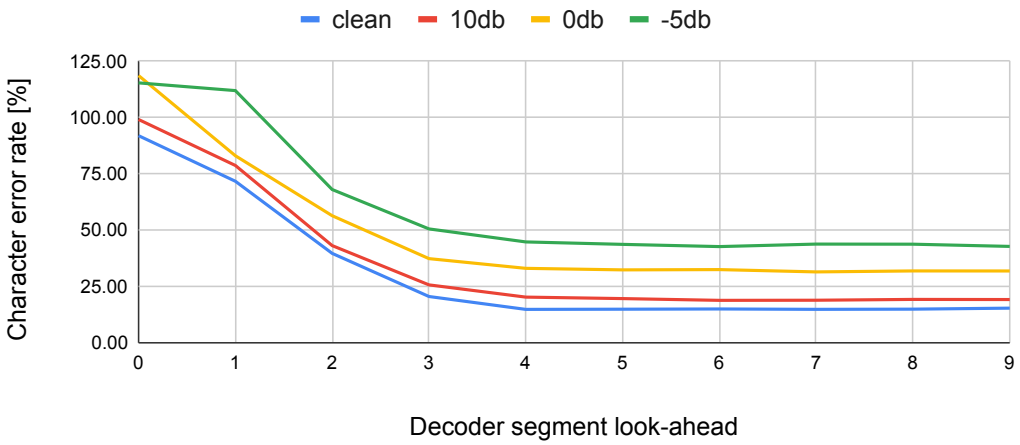


Figure 4: Online decoding performance on LRS2. We fix $e_{LB} = \infty$, $e_{LA} = 11$ frames, $d_{LB} = \infty$ and we only allow d_{LA} to vary.

We notice that there are diminishing returns after a context look-ahead d_{LA} of 4 words. Converted to audio frames, this value is higher on average than 330ms given by $e_{LA} = 11$ frames, as we will show in Figure 5. Thus, d_{LA} is the main indicator of the overall algorithmic latency given by the two quantities expressed in frames.

5.5 Evaluation on longer sentences

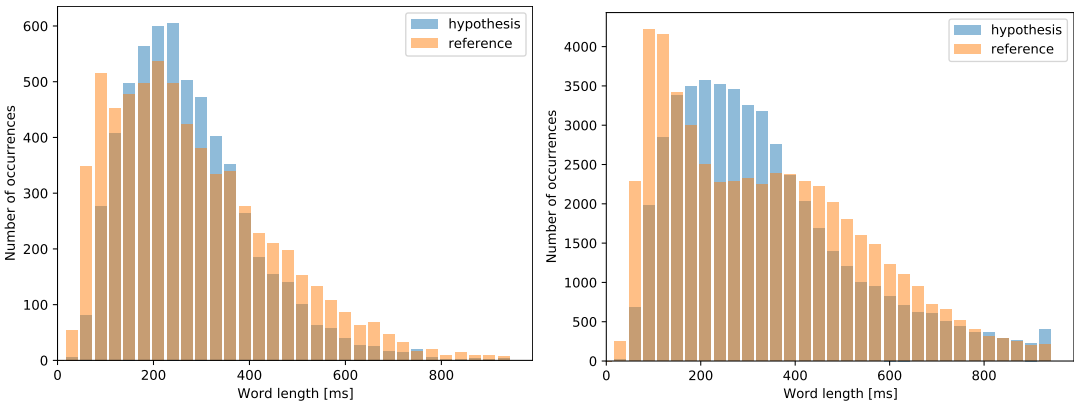
In the previous experiments we have used the LRS2 dataset for rapid prototyping. However, since it contains many short sentences, online decoding performance on the longer sentences risks being averaged out. We re-train and evaluate our models on the 100 hour clean partition of the LibriSpeech dataset, displaying the mean error and 95% confidence interval (CI) around the mean in Table 1.

First, we notice that the systems achieve an error rate similar to the one obtained on LRS2, despite the increased amount of data, suggesting that further gains are possible for larger model sizes. We also notice that the word loss can be slightly detrimental to the overall performance for the same network capacity, particularly for the models with unbounded attention span. This prompts a deeper investigation into the interplay between the cross entropy and word counting losses, as our constant scale factor λ is likely a less optimal solution to this multitask problem.

Table 1: System evaluation on LibriSpeech 100h clean partition

Model	parameters				**CER**		**Word Loss**
	e_{LB}	e_{LA}	d_{LB}	d_{LA}	mean [%]	95% CI [%]	
Transformer [38]	∞	∞	∞	∞	13.37	0.444	N/A
Transformer + Word Loss	∞	∞	∞	∞	14.64	0.451	0.92
Taris : infinite look-back	∞	11	∞	5	15.70	0.451	1.12
Taris : finite look-back	11	11	5	5	13.83	0.451	0.76

Additionally, we compare the distributions of the segment lengths estimated by Taris and those estimated with the pre-trained Montreal forced aligner [25]. We plot in Figure 5 the two length histograms. Not only are the histograms highly overlapped, but the one produced by Taris is in line with the average speaking rate of read speech. The small differences between the reference and hypothesis are likely owed to the short silences between words which were excluded from the reference, whereas Taris does not explicitly model silences and includes them into segments. Latency has not received sufficient consideration in prior work to facilitate a direct comparison, as systems



(a) LRS2 (b) LibriSpeech

Figure 5: Segmentation length distribution (in milliseconds) of Taris compared to the reference provided by the Montreal forced aligner

were trained with offline encoders [26] or large receptive fields [16], relied on beam search over the output distribution [11], or used phoneme units [17]. Very recent work combining a weaker online model with an offline rescorer [33] that allows to revise online hypotheses with a final hypothesis introduces the notion of end-pointing latency. The offline rescorer is triggered after the utterance has been determined to be finished, that is when a threshold period has elapsed after a suspected end of utterance without further speech activity. Since our model is fully online and does not have to wait to rescore, this metric is not applicable to our system.

6 Conclusion

We have proposed a simple, efficient, and fully differentiable solution for online speech recognition that does not require additional labels. Taris is inspired from early language acquisition in infants, and aims to segment a speech stream by learning to count the number of words therein. We show that our method matches the accuracy of an offline system once it listens to 5 dynamic segments. Lowering this latency remains a topic for exploration, e.g. by gradually reducing the look-ahead parameter d_{LA} later in training, explicitly modelling silences, or investigating the role of context, grammar, and semantics in lexical recognition.

Generalising to sentences of different lengths from the ones seen in training has been recently identified as a major problem for neural online speech recognition systems [7, 27]. By modelling only the local relationships in speech through finite look-back and look-ahead, we preserve the same property of the Neural Transducer [17] to effectively decouple the sentence length from the learnt representations, while allowing adaptive segments and simpler training.

It can be argued that Taris exploits human knowledge of the speech signal structure and embeds the concept of words and the local acoustic relationships, instead of being a more generic, self-organising neural network. Yet, the local processing of speech is merely the one dimensional equivalent of local convolutions applied to images, where the objects are replaced by words. Moreover, one-dimensional convolutions are commonly used in speech recognition [1, 29, 22, 21]. Given their major impact in research despite their lack of invariance to orientation, scaling, or even small perturbations, there is still much to be learned from engineered models in the pursuit of artificial general intelligence.

Broader Impact

Since our work only targeted the English language, it is unclear how Taris would perform on other languages with considerably different orthographies. The experiments on the train partition of LRS2 suggest that a relatively small dataset of unconstrained speech may be sufficient to allow the learning of typical segment length distributions. Given the local relationship modelling between input and output segments, we expect our solution to cope well with under-resourced languages.

Throughout the article we wanted to make a clear distinction between segments and words as acoustic units. Taris is not designed to align words just like a forced aligner would do, and should not be used for this purpose. The similarity between words and segments has not been thoroughly investigated in this study and is left as future work.

Acknowledgments and Disclosure of Funding

Our work is supported by a GPU grant from NVIDIA. The ADAPT Centre for Digital Content Technology is funded under the SFI Research Centres Programme (Grant 13/RC/2106) and is co-funded under the European Regional Development Fund.

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